SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Hokkien Mee Is Purple
Project / Solution Name	Varity
Name (Leader)	Tay Ee Zhan
Email & Phone (Leader)	On the signed copy sent to the organisers
University Affiliation(s)	On the signed copy sent to the organisers
Live Prototype URL	https://verity--eezhan0805.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/TayEeZhan/varity (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Varity is one Python server and a static web app. FastAPI, run with uvicorn on a Replit Autoscale deployment, serves the HTML, CSS and JavaScript screens and the JSON APIs, and imports the engine modules.

The engine reads every monthly Hacker News "Who is hiring?" thread through the Algolia HN Search API (January 2019 to October 2026, about 48,500 posts). DeepSeek V4 Flash, through the KrackedDevs gateway, names the roles in each post with a quote; code checks every quote. Code tags 40 skills, builds monthly shares, forecasts 12 months ahead with a backtested model and a conformal band, labels trends with a Welch t-test, and joins an AI-exposure score per skill. A ranker re-ranks AI news every 15 minutes.

Claude Haiku 5.5 points at CV skills with quotes and writes CV wording; code checks each answer. Accounts and profiles live in Replit PostgreSQL. Each data refresh is saved as a snapshot in Replit App Storage, and the app serves the newest good one. A GitHub Action starts the refresh every day at 09:00 Malaysia time.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Server, engine and tests	PSF License
JavaScript, HTML, CSS (no framework, no build step)	Web app in `frontend/`	Our own code
fastapi >=0.115; uvicorn[standard] >=0.30	Web server and JSON APIs	MIT; BSD-3-Clause
pydantic >=2.10	Checks API input and structured model output	MIT
anthropic >=2.0.0	Client for the Anthropic Messages API	MIT
httpx >=0.28; requests 2.34.2	HTTP clients (signal ranker; Algolia HN API)	BSD-3-Clause; Apache-2.0

Name & Version	Purpose in Project	License
apscheduler >=3.10,<4; feedparser >=6.0	Re-ranks AI signals every 15 minutes; reads RSS and Atom feeds	MIT; BSD-2-Clause
pypdf >=6.21	Reads text from an uploaded PDF CV	BSD-3-Clause
jinja2 >=3.1; markdown >=3.7; pyyaml >=6.0	Ranker pages, briefs, configuration	BSD-3-Clause; BSD-3-Clause; MIT
pandas 3.0.6; numpy 2.5.4; scipy 1.18.1; statsmodels 0.15.0	Tables, forecast maths, Welch t-test; damped trend (backtest only)	BSD-3-Clause (NumPy also 0BSD, MIT, Zlib, CC0)
psycopg[binary] 3.3.6; replit-object-storage 1.0.2	PostgreSQL client for accounts; Replit App Storage client (snapshots)	LGPL-3.0-only; ISC
typesafe-sdk 0.7.5	Optional Jev ranker (off by default)	MIT
pytest 9.1.1; pytest-cov >=5.0	1,375 automated tests	MIT
scikit-learn 1.9.1 (+ joblib, threadpoolctl)	Gradient-boosting experiment only, not deployed	BSD-3-Clause
Schibsted Grotesk and Lato (Google Fonts)	Web fonts	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Algolia HN Search API (hn.algolia.com/api)	Monthly "Who is hiring?" threads (job demand) and HN stories for the ranker	Free, public, no key
Anthropic Messages API (Claude Haiku 5.5; Claude Sonnet 5.5)	CV intake with quotes; route advisor; pathway coach; CV builder and CV assistant; project-jury seat; Signals story labels; optional Signals brief	Paid, on a team member's Anthropic account (Goh Sheng Kuan); key in Replit Secrets
New: KrackedDevs LiteLLM gateway (OpenAI-compatible)	DeepSeek V4 Flash: role labels on every HN post (48,491 posts, US$2.63 in total), project "why" lines, project-jury seat. GPT-5.4 mini: review of new job titles, project-jury seat	Sponsor-provided (KrackedDevs US$20 keys); key in Replit Secrets only
News and research sources: the RSS / Atom feeds in `backend/signals/config/sources.yaml`, the arXiv API, Hugging Face Daily Papers, the Bluesky public API	The AI-trend signal layer (Signals screen)	Free, public
Replit Auth (OpenID Connect with PKCE)	Sign-in	Part of the Replit app
New: GitHub REST API	Reads a public repository the user submits (owner, push days, files) for the project station	Free; server read-only token
New: Credly public badge pages	Checks a certificate badge the user submits	Free, public
DOSM open data (Excel download)	Malaysian online-vacancy check, sector level	Free, Government Open Data Terms of Use 1.0

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale deployment	Hosts the FastAPI server and the web app. Build `pip install -r requirements.txt`; run `python -m trendsignal serve --scheduler`	Autoscale, 2 vCPU / 4 GiB RAM, region Asia
Replit PostgreSQL	Accounts, saved profiles, CV drafts (SQLite file when no database is set)	Replit-hosted
New: Replit App Storage	Daily data snapshots (`latest.json` points to the newest good one; the last 14 are kept)	Default bucket of the app, set up 1 Nov 2026
Replit Secrets	`ANTHROPIC_API_KEY`, `KD_API_KEY`, `SESSION_SECRET`, `GITHUB_TOKEN`, `VERITY_STORE` and others; never in the repository	Replit
New: GitHub Actions	Daily refresh at 09:00 MYT: wakes the app, calls `POST /api/data/refresh`, waits until it ends (`.github/workflows/daily-refresh.yml`). Replit allows one deployment type per app, so a Replit Scheduled Deployment could not run beside the site	Free tier, ubuntu-latest runner

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

A language model never sets a forecast, score, rank, date or course. With no key, or over a daily budget cap, code takes over (keyword pass, formula ranking, code pathway).

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Haiku 5.5 (`claude-haiku-5-5`)	Anthropic API	Pre-trained. CV intake (points at skills with quotes; code checks each quote word for word); route advisor and pathway coach (re-order within code limits); CV builder and CV assistant (code rejects invented facts and stronger verbs); project-jury seat; Signals story labels	Anthropic commercial API terms
Claude Sonnet 5.5 (`claude-sonnet-5-5`)	Anthropic API	Pre-trained. Optional Signals brief writer; runs only when `TRENDSIGNAL_BRIEFS=1`	Anthropic commercial API terms
New: DeepSeek V4 Flash (`deepseek-v4-flash`)	DeepSeek, via the KrackedDevs gateway	Pre-trained. Names the roles in each HN job post with a quote (code drops a label whose quote is not in the post); writes the project "why" line (code checks it); project-jury seat	KrackedDevs sponsor terms; DeepSeek API terms
New: GPT-5.4 mini (`gpt-5.4-mini`)	OpenAI, via the KrackedDevs gateway	Pre-trained. Reviews new job titles before code adds a role (it needs three quotes from real posts); project-jury seat	KrackedDevs sponsor terms; OpenAI API terms
Forecast models: naive3 (mean of the last 3 months) in production; naive, drift and damped trend tested; conformal 80% band; Welch t-test	Our code (NumPy, SciPy, statsmodels)	Self-built statistical models, chosen by a rolling-origin backtest	Our code (Apache-2.0)

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Evaluation jury (offline, not in the running app): Claude Sonnet 5.5, GPT-6.1 Sol, Gemini 4 Argon	Anthropic, OpenAI, Google, via OpenRouter (Replit AI Integrations)	Pre-trained. Each labels the same news stories as human labellers, to grade the Signals ranker (`backend/signals/eval/jury.py`); NDCG@10 rose from 0.37 to 0.53 after the fixes it showed	OpenRouter and model-maker terms
HistGradientBoostingRegressor (scikit-learn 1.9.1)	scikit-learn	Self-trained experiment. **Not deployed:** it lost the backtest (2.06 vs 1.66 pp)	BSD-3-Clause

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Hacker News "Ask HN: Who is hiring?" threads, 2019-01 to 2026-10	hn.algolia.com/api	Public; we commit only aggregates and thread links	No data licence published; read through the public API only	Y in raw posts (kept out of the repository); N in committed files
Anthropic Economic Index, release 2026-06-26	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY (data)	N
GPTs are GPTs; AIOE; Microsoft Working with AI; O*NET 31.0	GitHub repositories and onetcenter.org (`data/SOURCES.md`)	Public (exposure cross-checks)	MIT; no licence file (cited); CC BY 4.0; CC BY 4.0	N
New: DOSM Big Data Analytics: Job Market Insights (quarterly)	dosm.gov.my	Public	Government Open Data Terms of Use 1.0	N
New: MyMahir Malaysia Critical Occupations List 2024/2025	TalentCorp (public PDF)	Public	Terms not reviewed	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
New: HRD Corp-claimable programmes (3, Trainocate Malaysia); project (11) and certificate (13) lists	Provider pages, checked 31 Oct 2026 (`data/hrd_programmes.json`, `data/projects.json`, `data/certs.json`)	Self-collected (titles, days, fees, links)	Providers' own terms	N
Learning resources: 77 links (76 free, 1 free trial), HTTP 200 on 22 Oct 2026	`data/resources.json`	Self-collected (titles and links)	Providers' own terms	N
Example profiles (Aina, Ben, Chloe) and 7 demo CVs	`data/example_profiles.json`, `demo-cvs/`	Synthetic	Our own work	N (synthetic)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic CLI), with Claude Opus 5.5, Sonnet 5.5 and Haiku 5.5, and its skills and plugins (for example design and code-review skills)	Code, tests, data pipeline, documents, UI design and the CodeTour guides (`.tours/`), from the team's prompts and plans. 126 commits carry its "Co-Authored-By: Claude" line; `CLAUDE.md` and `.claude/` hold its project settings	Major
Replit Agent	Replit environment setup: the `.replit` file, the App Storage bucket link and the "Published your App" commits (27). It wrote no app code	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Both members read and run every change, and every pull request is reviewed. 1,375 automated tests run before a merge, including tests that block look-ahead leaks and tests that check every model answer. Every statistic traces to a file in `data/` or a cited source. Reviews found real faults that we fixed: a bare "AI" keyword inflated Machine Learning to 55% of posts, the first exposure score measured traffic, and the first uncertainty band covered only 65.9%. Both members can explain each part.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
KrackedDevs	US$100 AI API credit (five US$20 keys), LiteLLM gateway	DeepSeek V4 Flash labelled the roles in 48,491 HN posts (US$2.63, `data/llm_usage.csv`) and labels each day's new posts; GPT-5.4 mini reviews new job titles; both sit on the project jury
Replit	Platform: Autoscale deployment, Auth, PostgreSQL, App Storage, Secrets	Hosts the live app, sign-in, accounts and the data snapshots. The US$100 Replit credits did not reach us; the organisers waived the Replit proof of usage
RunPod	None	Not used: no GPU work

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Both can be replaced. KrackedDevs makes the role labels sharper and cheap; without its key, code falls back to title rules or to Claude. Replit hosts the app; the server is plain FastAPI, the snapshot store has a local-folder version (`engine/store.py`) and the daily job is one command, so any Python host works.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Users sign in with Replit Auth. We store the Replit user id, email, display name and profile image link, which are personal data under Malaysia's PDPA 2010, only to keep the user's profile and progress. A profile holds skills with their evidence quotes, the target role, the plan and its ticks, and, for the CV builder, the cleaned CV text. Code replaces emails, phones and links with placeholders before any model call and before storage. We never store the uploaded file. Data sit in Replit PostgreSQL; we sell or share none of it. A user can delete the account at any time, and an account unused for 365 days is deleted. CV text goes to Anthropic (Claude Haiku) for reading. Only public Hacker News text goes to KrackedDevs. Real CVs are out of scope: the consent box asks for synthetic CVs. Privacy page: `frontend/privacy.html`.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Hacker News is a US and startup sample, not Malaysian; the app shows a DOSM sector check beside it.
2. Twelve months ahead, persistence beats trend models, so most roles read "Steady"; we show a band, not a promise.
3. The Fast scenario's k = 2 is an assumption (the data interval is −1.0 to 2.9).
4. A model can miss or invent; every answer is checked against quotes or code limits, and code takes over on failure.
5. Single providers (Anthropic, KrackedDevs, Replit): each has a code fallback and a daily budget cap.
6. Autoscale sleeps: the daily refresh wakes it, and a failed run keeps the last good snapshot.
7. The fairness tests cover name, age, gender, nationality and school, not every bias.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: signed copy submitted to the organisers Name: Tay Ee Zhan (team leader)

Date: 1 November 2026